

384.185 Machine Learning Projects



Project

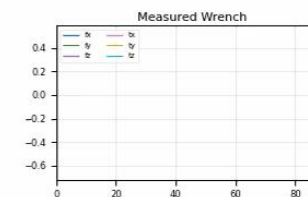
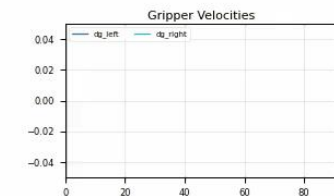
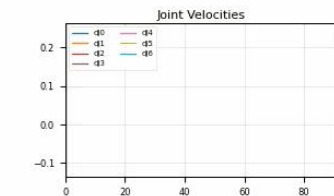
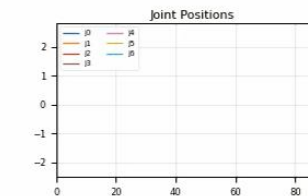
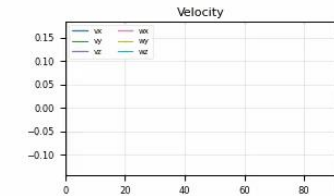
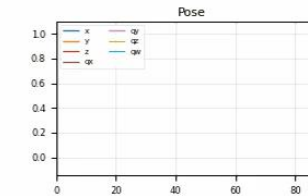
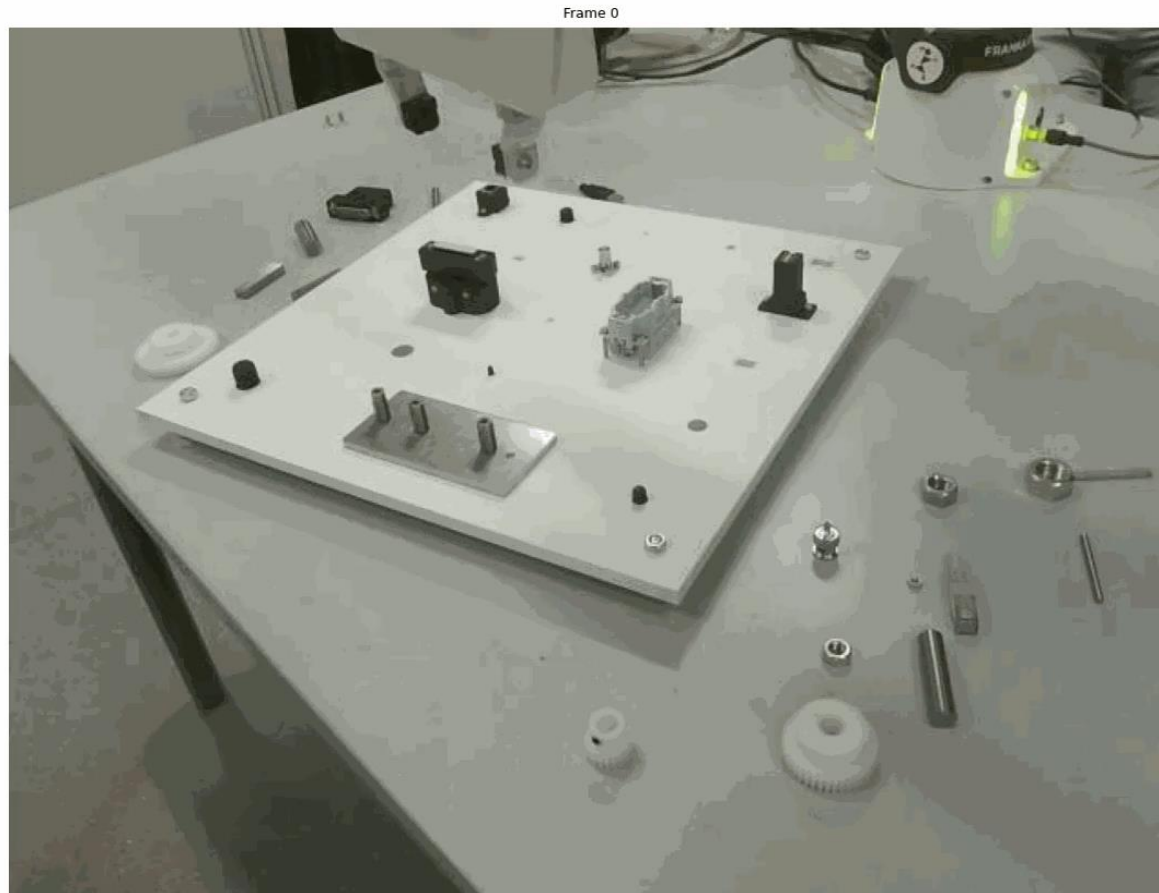
- The Project counts for 40 points
 - 10 points are obtained from a Report
 - 30 points are obtained from task performance
- Your goal is to solve 2 machine learning problems:
 - Activity Recognition
 - Temporal Action Segmentation
 - (Optionally): Anomaly Detection
- You will use a subset of the REASSEMBLE dataset:
 - Daniel Sliwowski, Shail Jadav, Sergej Stanovcic, Jedrzej Orbik, Johannes Heidersberger, Dongheui Lee, Demonstrating REASSEMBLE: A Multimodal Dataset for Contact-rich Robotic Assembly and Disassembly, Robotics: Science and Systems (RSS), pp. -, 2025 [Project Page](#) | [ArXiV Page](#)
- Your Report should be about 1 page in length (max. 2 pages)

Dataset

- The considered subset of the REASSEMBLE dataset has 4 actions and 3 objects:
 - Pick, Insert, Remove, Place
 - Small, Medium, and Large gear
- It contains demonstrations of a robot performing a gear assembly and disassembly task.
- There are 3 splits:
 - Training: 98 demos
 - Validation: 18 demos
 - Test: 21 demos (you do not have access to these)
- Access to data:
 - Train: https://drive.google.com/file/d/1idQ60dI_2AUK5oIKvHuPWLzr3Mf5Jnjp/view?usp=sharing
 - Validation: <https://drive.google.com/file/d/1STRwITAHSOn9q2n-7AJ1gTvENaUuO1tn/view?usp=sharing>
- Each demonstration contains recordings from various sensors all recorded at 10 FPS:
 - Camera
 - End effector pose
 - End effector twist
 - End effector wrench
 - Gripper width
 - Joint positions
 - Joint velocities

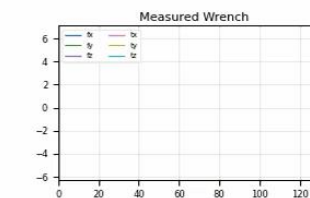
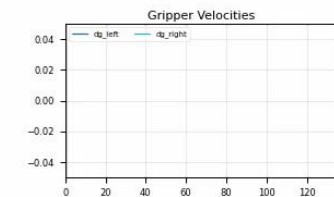
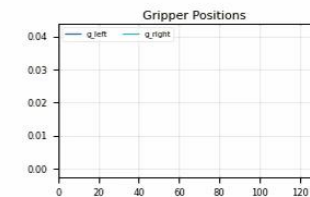
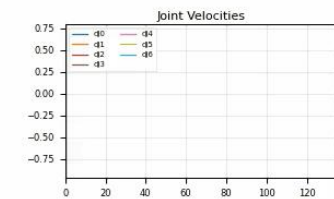
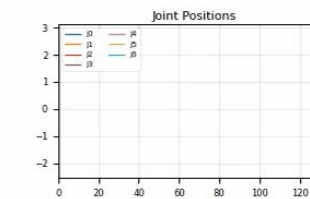
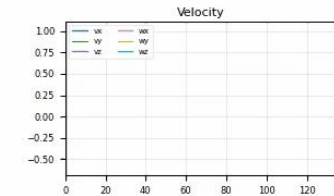
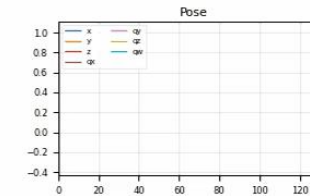
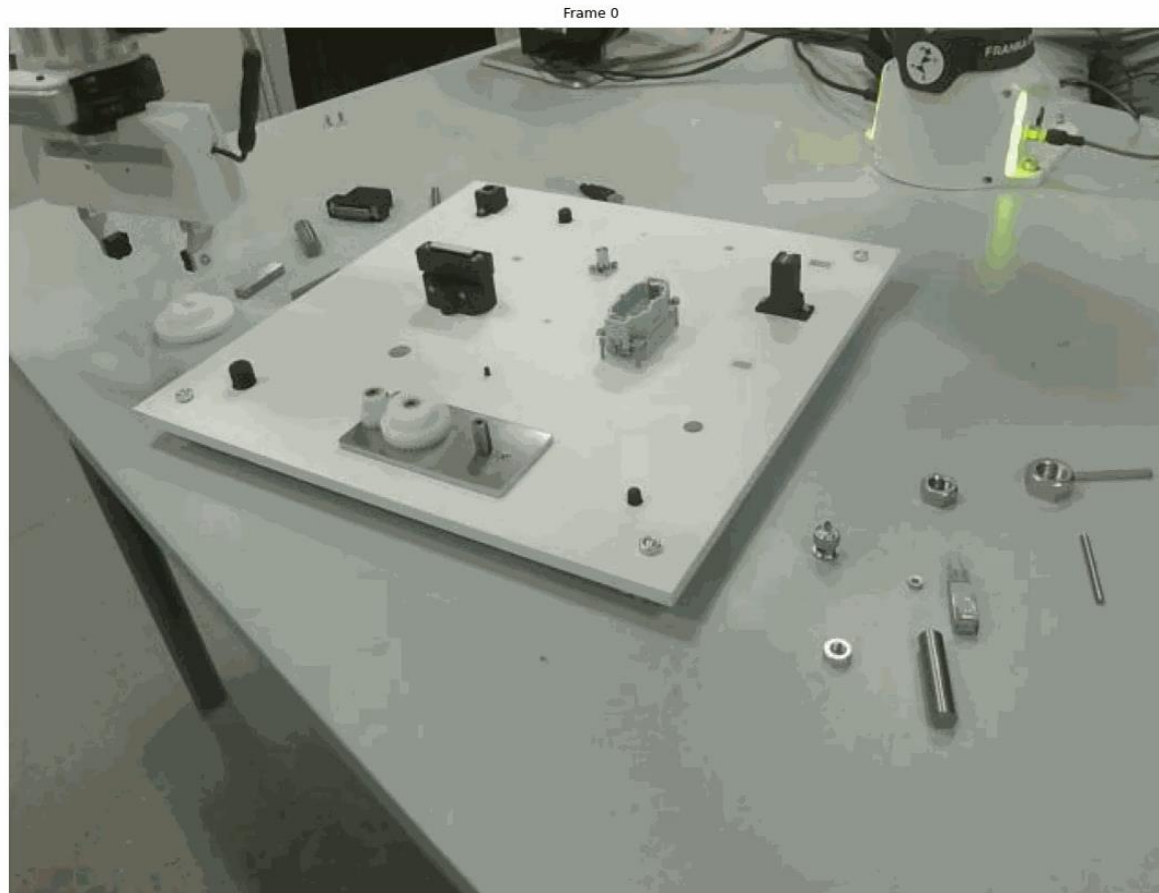
Dataset: Action Recognition

Successful Sample | Action: pick medium gear | Success: ✓



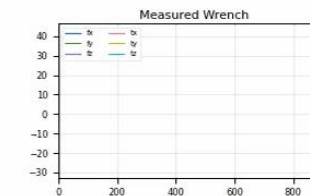
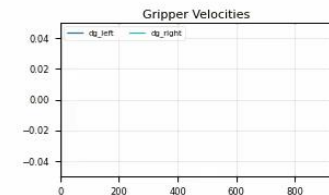
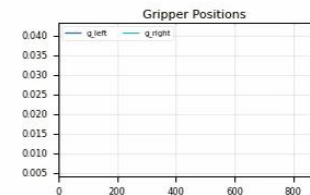
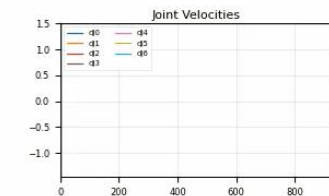
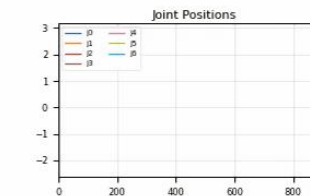
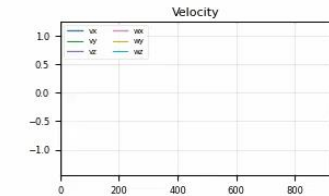
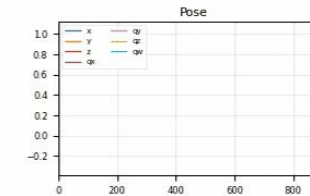
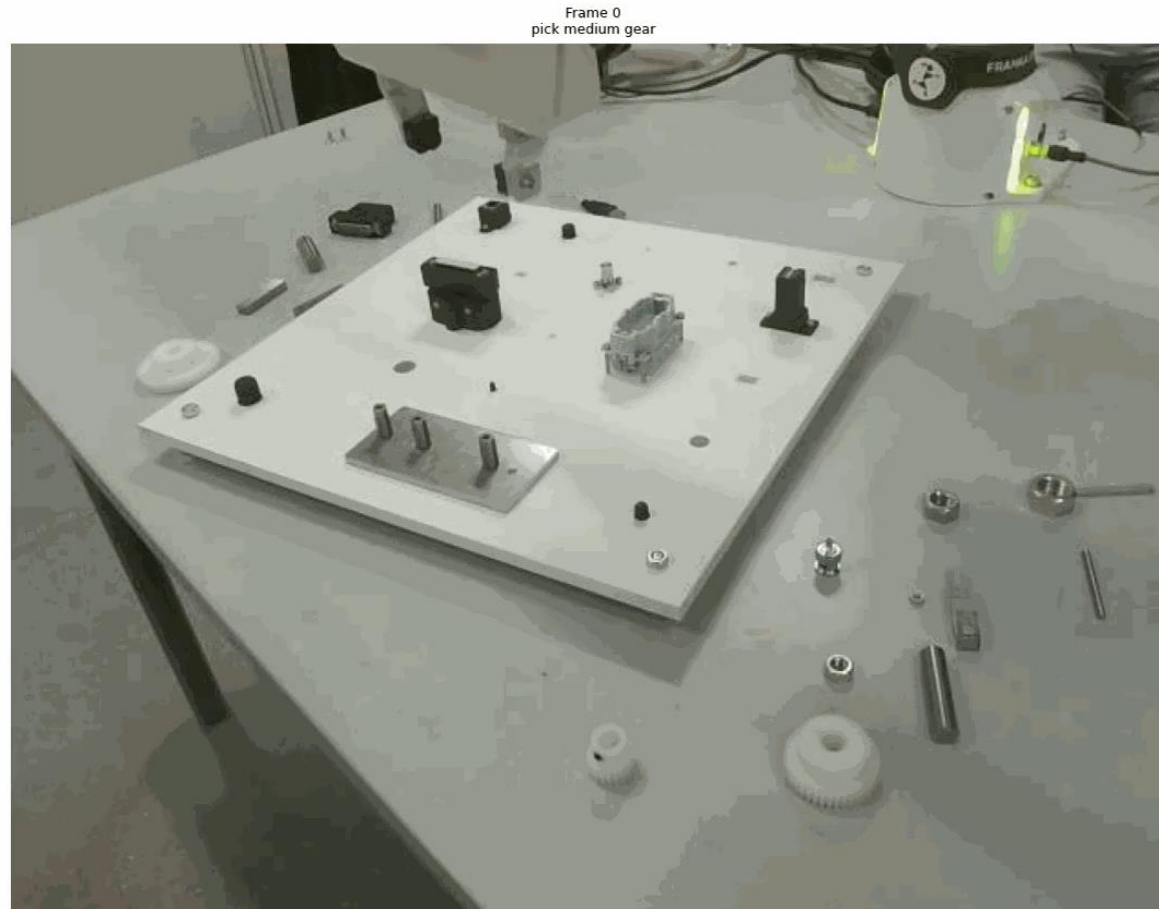
Dataset: Action Recognition

Failed Sample | Action: remove medium gear | Success: $\times$



Dataset: Temporal Action Segmentation

Segmentation Sample



Action Recognition: Formulation

Input

Trimmed clip $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ of N sequential observations, where each $\mathbf{x}_t \in \mathbb{R}^d$ is a feature vector of dimension d

Choice of features determines d . You are free to select any features from the dataset.

Output

Predicted action class $\hat{y} \in \mathcal{Y} = \{0, 1, 2, \dots, 12\}$

Each integer corresponds to a distinct action category ($|\mathcal{Y}| = 13$)

Model

Classification function $f_{\theta} : \mathbb{R}^{N \times d} \rightarrow \mathcal{Y}$ parameterized by θ

Goal

Find optimal θ^* minimizing expected misclassification over data distribution $\mathcal{D}$

Action Recognition: Evaluation

All metrics use micro-averaging: compute globally by summing TP, FP, FN across all 13 classes before dividing.

Accuracy

$$\text{Accuracy} = \text{correct} / \text{total}$$

Fraction of clips correctly classified out of all clips.

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Of all clips predicted as a class, how many truly belong to it.

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Of all clips that truly belong to a class, how many were found.

F1 Score

$$\text{F1} = 2 \cdot \text{TP} / (2 \cdot \text{TP} + \text{FP} + \text{FN})$$

Harmonic mean of Precision and Recall — balances both.

TP = True Positives · FP = False Positives · FN = False Negatives · Micro-averaging treats the full dataset as one pool.

Temporal Action Segmentation: Formulation

Input

Untrimmed data stream $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ of N sequential observations, where each $\mathbf{x}_t \in \mathbb{R}^d$ is a feature vector of dimension d
Choice of features (e.g. RGB pixels, optical flow, skeleton keypoints) determines d

Output

Label sequence $\hat{\mathcal{Y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ of length N
where each $\hat{y}_t \in \mathcal{Y} = \{0, 1, \dots, 12\}$ is the predicted action class at time step t

Model

A sequence labelling function $f_{\theta} : \mathbb{R}^{N \times d} \rightarrow \mathcal{Y}^N$ parameterized by θ

Goal

Find θ^* minimizing frame-level misclassification over distribution $\mathcal{D}$

Temporal Action Segmentation: Evaluation

Accuracy

$$\text{Accuracy} = \text{correct} / \text{total}$$

Fraction of frames correctly labelled across all frames in all videos.

Edit Score

$$\text{Edit} = (1 - \text{edit_distance}(P, Y) / \max(|P|, |Y|)) \times 100$$

P, Y = segment-level sequences (consecutive duplicate labels collapsed). edit_distance counts insertions, deletions, substitutions.

F1@k

$k = 0.10$

$k = 0.25$

$k = 0.50$

k is the IoU threshold for a predicted segment to count as a True Positive.

TP

A predicted segment is TP if $\text{IoU} \geq k$ with a same-label GT segment (each GT matched at most once).

FP

Predicted segment with $\text{IoU} < k$ for every GT segment of the same label.

FN

Ground-truth segments that were not matched by any predicted segment.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1@k} = 2 \cdot \text{TP} / (2 \cdot \text{TP} + \text{FP} + \text{FN}) \times 100$$

IoU

$$\text{IoU} = (\min(p_end, y_end) - \max(p_start, y_start)) / (\max(p_end, y_end) - \min(p_start, y_start))$$

For same-label segments: predicted $[p_start, p_end)$ vs. ground-truth $[y_start, y_end)$

Anomaly Detection: Formulation

Trimmed clip $\rightarrow$ binary outcome (success / anomaly)

Input

Trimmed clip $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ of N sequential observations, where each $\mathbf{x}_t \in \mathbb{R}^d$ is a feature vector of dimension d

Choice of features (e.g. RGB pixels, optical flow, skeleton keypoints) determines d

Output

A binary label $\hat{y} \in \mathcal{Y} = \{0, 1\}$ where:

0 $\rightarrow$ Anomaly (failed demonstration)

1 $\rightarrow$ Success (correct demonstration)

Model

A binary classifier $f_{\theta} : \mathbb{R}^{N \times d} \rightarrow \{0, 1\}$ parameterized by θ

Goal

Find θ^* minimizing misclassification — prioritizing detection of anomalies over a distribution $\mathcal{D}$

Anomaly Detection: Evaluation

Positive class ($\hat{y} = 1$): Success · Negative class ($\hat{y} = 0$): Anomaly · TP/FP/FN defined with respect to the positive (success) class.

Accuracy

$$\text{Accuracy} = \text{correct} / \text{total}$$

Fraction of clips correctly classified (success or anomaly).

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Of all clips predicted as success, how many truly were.

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Of all true successes, how many were correctly predicted.

F1 Score

$$\text{F1} = 2 \cdot \text{TP} / (2 \cdot \text{TP} + \text{FP} + \text{FN})$$

Harmonic mean of Precision and Recall — key metric when classes are imbalanced.

Deliverables

Code:

1. You will be required to upload your entire codebase for us to evaluate.

Report:

1. Clearly present the approaches you used to solve the project, where you explain all assumptions and design decisions.
2. Clearly present and discuss the performance of the developed approaches.
3. Analyze the impact of your design decisions on the performance of the model.
4. Include both quantitative (metrics) and qualitative (images) results.
5. If you base your approach on any existing works please briefly explain and cite said works.

Leaderboard Submissions:

1. Your Task Performance will be determined based on the evaluation on a left out test set to which you will not have access. We are still looking into how best to organize this process and will announce how to submit your results soon.

Codebase

You can obtain code for loading the data and computing the metrics here:

- https://github.com/TUWIEN-ASL/MachineLearning_384185_Project

Thank you for listening



*Autonomous
Systems Lab*

